SubterraView

Group 4

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Spring 2026

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I Project Description

1 Project Overview

SubterraView is a system that employs Augmented Reality/Artificial Intelligence (AR/AI) to map and visualize underground infrastructure in real time. The system creates a continuously updated map of subterranean infrastructure such as plumbing, telephone cables, and power lines data from compiling public utility records, city databases, and crowdsourced reports. An AI engine cross-references data with each other to validate the information, ranking specifics via reliability scores to resolve information conflicts and retain accuracy. The map is generated through an AR system that allows users to visualize infrastructure directly on their smartphones.

The system is meant to provide service to construction workers, utility technicians, and emergency responders who currently rely on the "811 call-before-you-dig" process, a process that introduces significant project delays and opens the possibility of infrastructure damage from human error. SubterraView incorporates a hazard system that provides real-time alerts to workers and systems of dangerous conditions such as gas leaks and structural instabilities, things that are picked up from our portable sensing technology to detect dangers that may go unnoticed by human workers. A crowdsourcing platform allows field workers to report discrepancies, new utilities, and infrastructure conditions–things that could get lost when it comes to human filing– which the AI model uses to continuously update the map.

2 The Purpose of the Project

2a The User Business or Background of the Project Effort

The underground utility and infrastructure management industry supports the installation, maintenance, and repair of underground infrastructure including water and sewage pipelines, cables, and electrical conduits across cities. Currently, contractors must contact 811, a national call before you dig service, to have utility lines marked before any excavation can begin, a process that is time consuming and heavily reliant on outdated records. According to the Common Ground Alliance, nearly 197,000 unique underground utility damage incidents were recorded in 2024 alone [1]. SubterraView is intended to give construction workers, utility technicians, and emergency responders a real time, AI validated map of underground infrastructure accessible directly from their smartphones, eliminating the need to rely solely on the 811 processes.

2b Goals of the Project

The main goal of this project is to develop a reliable and accessible underground infrastructure AI mapping system that enables real time visualization that's cross-referenced with current validated records, and hazard monitoring for workers operating in subsurface environments. Some of the objectives are to streamline the current notification workflow, minimize accidental utility strikes and the financial damages they cause, provide real time hazard alerts for dangerous conditions (such as gas leaks and structural instabilities), and empower field workers to contribute and access accurate infrastructure data directly from their smartphones.

2c Measurement

- 30 percent or greater reduction in accidental utility strike incidents among active SubterraView users compared to the national average
- Reducing pre-excavation preparation time from the current construction and excavation industry average of several days to 24 hours or less
- AI map accuracy rate reaches a minimum of 90 percent when validated against verified and up-to-date utility records
- A reduction in worker injury rates in subsurface environments where the hazard monitoring system is deployed by determining 85% of hazard alerts must be acted on within 5 minutes of being issued

3 The Scope of the Work

The business environment for SubterraView is municipal underground infrastructure management and safety coordination. The work addressed by this project is the digital detection, validation, visualization, and hazard monitoring of underground infrastructure data to support construction planning and emergency response.

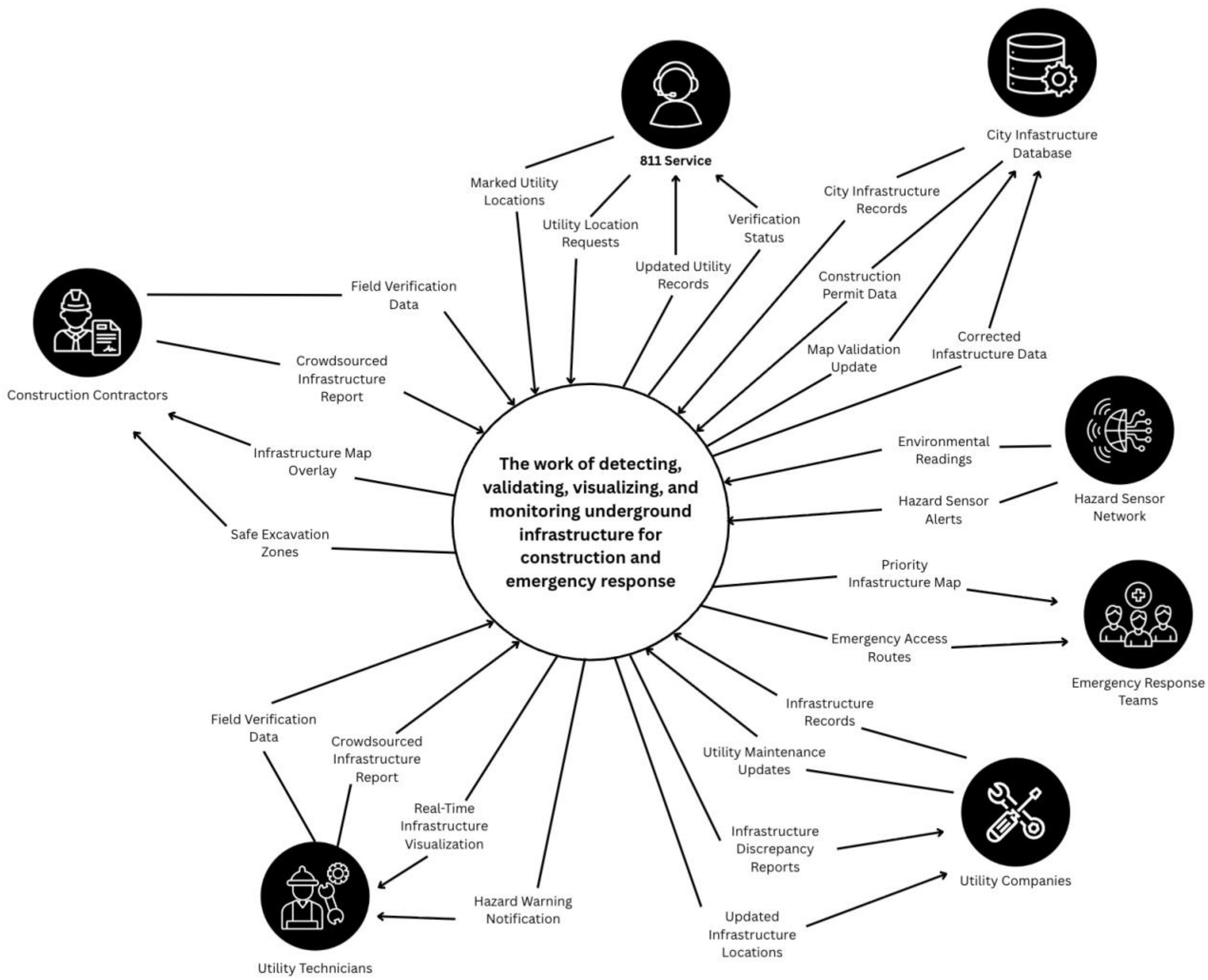
3a The Current Situation

Currently, underground infrastructure management relies on a combination of 811 notification systems, fragmented city and utility databases, surface paint marking, and manual field verification. Before excavation, construction companies must submit requests to 811 services, which notify relevant utility providers. Utility technicians then visit the site to physically mark underground utilities. Not to mention, infrastructure data is often stored across multiple systems maintained by different utility companies which can cause discrepancies in format and inconsistent update frequencies. Emergency responders typically rely on existing city maps or utility contacts and hazard detection systems such as gas sensors, operate independently and aren't always on the mapping platforms. Overall, the process is time consuming and inconsistent. Delays in approval, outdated records, and inconsistent data increase the risk of accidental infrastructure damage and worker injuries.

3b The Context of the Work

The work context diagram below defines the boundary of underground infrastructure management work that SubterraView supports. The central work area encompasses the digital detection, validation, visualization, and hazard monitoring of underground infrastructure data. External entities include 811 services, utility companies, city infrastructure databases, construction contractors, utility technicians, emergency response teams, and hazard sensor networks. Each interface represents a critical data exchange required to support construction planning, excavation safety, and emergency response coordination. The diagram shows both input flows such as infrastructure records, excavation requests, and hazard alerts, and output flows such as infrastructure map overlays, priority emergency maps, and safety notifications. This context establishes the scope of work analysis required to ensure SubterraView integrates seamlessly into the existing underground infrastructure management ecosystem.

Figure 1 – Interaction Diagram



3c Work Partitioning

Event Name	Input and output	Summary
Construction Request Initiated	Excavation Location Request (in), Project Site Coordinates (in), Infrastructure Map Overlay (out), Safe Excavation Zones (out)	When a construction project begins, the system retrieves and displays underground infrastructure data for the selected area.

811 Service Request Received	Utility Location Requests (in), Marked Utility Locations (in), Verification Status (out), Updated Utility Records (out)	When an 811 request is received, the work processes utility location data and provides verification status back to the service.
Utility Data Updated	Infrastructure Records (in), Utility Maintenance Updates (in), Updated Infrastructure Locations (out), Infrastructure Discrepancy Reports (out)	When new or modified utility records are received, the system validates and integrates the data into the map and reports any discrepancies.
Hazard Detected	Hazard Sensor Alert (in), Environmental Readings (in), Hazard Warning Notification (out)	When abnormal underground conditions are detected, the system alerts workers and relevant authorities.
Crowdsourced Report Submitted	Crowdsourced Infrastructure Report (in), Field Verification Data (in), Map Validation Update (out), Corrected Infrastructure Data (out)	When a worker reports a discrepancy or new infrastructure, the system verifies and updates the model.
Emergency Response Initiated	Emergency location query (in), Priority Infrastructure Map (out), Emergency Access Routes (out)	When emergency responders request information, the system provides a focused underground layout and safe access routes.
Permit Data Retrieved	Construction Permit Data (in), City Infrastructure Records (in), Infrastructure Map Overlay (out)	When construction permit data is received from city databases, the system incorporates it into the infrastructure visualization.

3d Competing Products

Several existing systems address underground infrastructure mapping and safety coordination, but only partially. The most widely used alternative is the 811 service, which notifies utility providers before excavation begins. While effective for basic coordination, this process is manual, time consuming, and does not provide real-time visualization. There are also other products such as Geographic Information System (GIS) and Ground-Penetrating Radar (GPR), but they are typically static, fragmented, and don’t integrate data into a centralized infrastructure model. Although these solutions provide partial functionality, none combine AI-based validation, AR visualization, real-time updates, and integrated hazard detection into a unified system. SubterraView aims to address these gaps by offering a comprehensive, underground infrastructure platform.

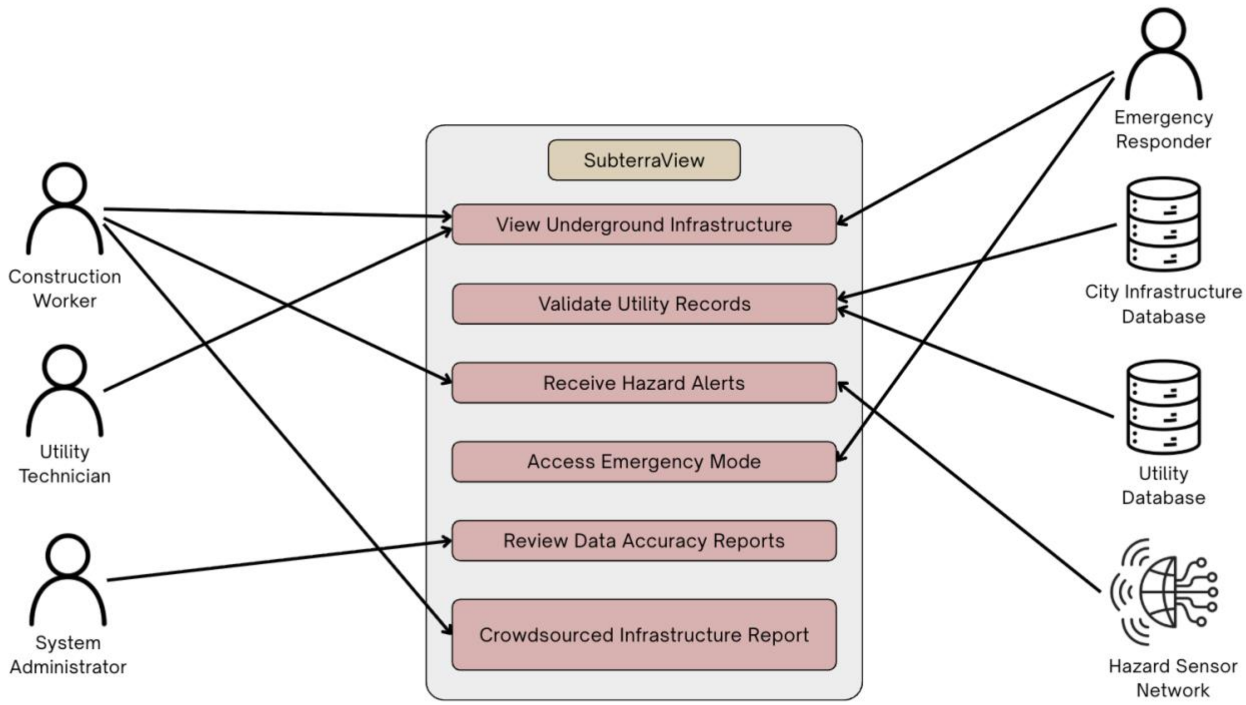
4 The Scope of the Product

SubterraView handles the digital detection, validation, visualization, and hazard monitoring of underground infrastructure data. The product does not handle physical excavation, regulatory enforcement, or long-term infrastructure maintenance. These remain the responsibility of external organizations. The scenarios below demonstrate how construction workers, utility technicians, emergency responders, and system administrators would interact with SubterraView in real-world situations.

4a Scenario Diagram(s)

The scenario diagram below illustrates the scope of the SubterraView product and the interactions between external actors and system functions. The diagram shows three types of users on the left: construction workers, utility technicians, and system administrators. On the right are the external systems that provide data to or receive data from SubterraView: emergency responders, city infrastructure database, utility database, and the hazard sensor network. The central box contains the core functions of the product including viewing underground infrastructure, validating utility records, receiving hazard alerts, accessing emergency mode, reviewing data accuracy reports, and submitting crowdsourced infrastructure reports. Each arrow represents a specific interaction between an actor and a system function, with the complete set of interactions mapping to the individual product scenarios described in section 4c.

Figure 2 - Use Case/Scenario Diagram



4b Product Scenario List

Scenario Name	External Actors Involved
1. AR Underground Visualization	Construction Worker
2. Utility Data Update and Validation	Utility Database, City Infrastructure Database
3. Crowdsourced Infrastructure Report	Utility Technician
4. Hazard Detection and Alert Notification	Hazard Sensor Network, Construction Worker, Emergency Responder
5. Emergency Infrastructure Access	Emergency Responder
6. Periodic Data Accuracy Review	System Administrator

4c Individual Product Scenarios

Scenario 1: AR Underground Visualization

A construction worker arrives at a new job site to begin excavation. Instead of waiting for surface painting markings or contacting multiple utility providers, the worker opens the SubterraView application on their smartphone. After selecting the project location, the app overlays underground infrastructure directly onto the camera view, showing water lines, gas pipes, electrical conduits, and telecommunications cables beneath the surface. Each utility line is color-coded and labeled for clarity. The worker adjusts the viewing angle and zooms level to better understand the layout and identifies a safe excavation path. This allows the team to begin work confidently and efficiently without unnecessary delays.

Scenario 2: Utility Data Update and Validation

A utility company completes an upgrade to a section of underground electrical wiring and updates their records in the utility database. The updated infrastructure records are automatically submitted to SubterraView. The system integrates the new data into its centralized model and cross-references it with existing records from the city infrastructure database. If conflicting information is detected, the system flags the discrepancy for review and assigns a confidence rating to the updated section. Once validated, the infrastructure map is updated so that future construction crews and responders have access to the most accurate information available.

Scenario 3: Crowdsourced Infrastructure Report

While working underground, a utility technician discovers a cable that is not shown on the digital map. Using the SubterraView app, the technician submits a report including the location, description, and supporting images. The system records the submission and compares it with existing infrastructure data and previous reports. After review, the infrastructure model is updated, and the confidence rating for that area is adjusted accordingly. This ensures that the system continuously improves and reflects real-world conditions.

Scenario 4: Hazard Detection and Alert Notification

During excavation, a connected underground gas sensor detects abnormal methane levels. The hazard monitoring system immediately sends a warning notification to the construction worker through the SubterraView application. The AR interface highlights the affected underground area in red. Workers evacuate the area immediately and emergency responders are notified of the hazard location to prevent potential injuries or explosions.

Scenario 5: Emergency Infrastructure Access

An underground structural collapse occurs during a construction project. Emergency responders arrive at the scene and access SubterraView in emergency mode. By entering the location, responders instantly view a prioritized underground map showing utility lines, confined spaces, and possible hazards. This allows the response team to determine the safest entry points and identify potential danger zones before beginning rescue operations.

Scenario 6: Periodic Data Accuracy Review

At scheduled intervals, a system administrator initiates a data integrity review of stored infrastructure records. The system analyzes the map and flags sections with outdated or low confidence data for further inspection. The administrator receives a detailed report identifying areas that may require updated verification or additional field data. Using this report, the administrator coordinates follow-up actions to ensure the infrastructure map remains accurate and reliable over time.

5 Stakeholders

5a The Client

The primary client for SubterraView is municipal public works departments. These are government agencies responsible for managing and maintaining underground infrastructure including water, sewer, and electrical systems across cities. Municipal public works departments have a direct financial and operational interest in reducing utility damage incidents and improving excavation safety coordination within their jurisdictions.

5b The Customer

The primary customers for SubterraView are construction companies and utility companies. Construction companies represent the largest end-user segment in the underground utility mapping market, driven by the need for accurate subsurface data before excavation begins. Utility companies are also key customers, as they rely on accurate infrastructure mapping to manage, maintain, and update their underground networks. Emergency response agencies represent a secondary customer group that would adopt SubterraView for hazard coordination purposes.

5c Hands-On Users of the Product

1. Construction Worker:

    (a) User role: Uses SubterraView in the field to view underground infrastructure before and during excavation, reducing the risk of accidental utility strikes.

    (b) Subject matter experience: Journeyman to Master, as familiarity with underground infrastructure on the field is expected.

    (c) Technological experience: Novice to Journeyman, based on general comfort with mobile technology.

(d) Other user characteristics: Field based workers who need quick and reliable access to data while on site. May work in loud or physically demanding environments where ease of use is important.

2. Utility Technician

    (a) User role: Uses SubterraView to verify infrastructure records, submit crowdsourced reports, and flag discrepancies found during field work.

    (b) Subject matter experience: Master, as utility technicians are expected to have strong knowledge of underground systems and infrastructure.

    (c) Technological experience: Journeyman, as comfort with digital reporting tools is expected in this role.

    (d) Other user characteristics: Often works independently in the field and relies on accurate data to complete maintenance and inspection tasks safely.

3. Emergency Responders:

    (a) User role: Uses SubterraView in emergency mode to access real time underground infrastructure maps during active incidents such as structural collapses or gas leaks.

    (b) Subject matter experience: Master, as emergency responders are expected to operate quickly and accurately under pressure.

    (c) Technological experience: Journeyman to Master, as above average comfort with technology is necessary when responding to time sensitive emergencies.

    (d) Other user characteristics: Response needs will vary depending on the type of underground emergency. Fast load times and a clear interface are critical for this user group.

4. Household Individuals:

    (a) User role: Able to use SubterraView for personal projects allowing for better planning.

    (b) Subject matter experience: Novice to Journeyman, as some basic knowledge of underground aspects may be required.

    (c) Technological experience: Novice to Journeyman, comfortable with mobile apps and understanding basic visualizations.

    (d) Other user characteristics: Use cases will vary from casual app usage to hands-on personal projects. Basic knowledge regarding underground aspects should be known.

5d Maintenance Users and Service Technicians

1. Software Engineers and AI Engineers:

    (a) Responsible for deploying updates, resolving software bugs, and improving core features such as the AR visualization engine and AI validation model to keep SubterraView functional and accurate over time.

2. IT Support Technicians:

    (a) Description: Responsible for resolving hardware compatibility issues, managing device support requests, and assisting users who experience technical difficulties with the application.

3. Database Administrators:

    (a) Responsible for maintaining the integrity and security of stored infrastructure records, managing data pipelines from external sources such as the City Infrastructure Database and Utility Database, and ensuring data is properly formatted and accessible.

5e Other Stakeholders

1. Testers:

    (a) Knowledge needed: Software testing experience to identify bugs and performance issues, as well as working knowledge of underground infrastructure to validate that the app functions accurately in real world conditions.

    (b) Degree of involvement: High, since testers will be involved throughout the development cycle, providing structured feedback and verifying that issues are resolved before deployment.

    (c) Degree of influence: High, since their findings directly affect whether SubterraView is approved for release.

2. Legal Experts

    (a) Knowledge needed: Knowledge of federal and state regulations governing excavation safety, including 811 compliance requirements, to ensure SubterraView operates within all applicable legal boundaries.

    (b) Degree of involvement: Low, since legal experts will be consulted during the early stages of development and when significant changes to the product are made.

(c) Degree of influence: High, since SubterraView cannot be deployed without confirming that it complies with all relevant regulations.

(d) Conflict resolution: If legal requirements conflict with planned features, legal compliance takes precedence and the feature will be modified or removed accordingly.

3. Sponsors

(a) Knowledge needed: General understanding of the underground infrastructure management industry and the value SubterraView provides to construction and utility operations.

(b) Degree of involvement: Low, since sponsors are primarily involved during funding decisions and major development milestones.

(c) Degree of influence: High, since sponsor funding directly affects the pace and scope of development.

(d) Conflict resolution: If sponsor expectations conflict with technical or legal requirements, the development team will communicate constraints clearly and propose adjusted timelines or feature scopes.

5f User Participation

1. Construction Worker:

(a) Description: Construction workers will contribute field level feedback on the accuracy of the AR infrastructure overlay and the usability of the interface in active job site conditions. Their input will be essential for validating that the system performs reliably during excavation planning.

(b) Time contribution: Involved during initial testing phases and on an ongoing basis after deployment through the crowdsourced reporting feature.

2. Utility Technician

(a) Description: Utility technicians will provide subject matter expertise on infrastructure data accuracy and help validate that crowdsourced reporting functions correctly in the field. Their feedback will be used to refine the confidence rating system.

(b) Time contribution: Involved during testing phases, particularly for scenarios related to data validation and discrepancy reporting.

3. Emergency Responders:

(a) Description: Utility technicians will provide subject matter expertise on infrastructure data accuracy and help validate that crowdsourced reporting functions correctly in the field. Their feedback will be used to refine the confidence rating system.

(b) Time contribution: Involved during testing phases, particularly for scenarios related to data validation and discrepancy reporting.

5g Priorities Assigned to Users

1. Key users: Construction Worker and Utility Technician - 70%

    (a) Construction workers and utility technicians are the primary users of SubterraView and represent the core use cases the product is designed to support. Their day-to-day reliance on accurate infrastructure data means their requirements take the highest priority in design and development decisions. Feedback from these users has the greatest influence on product iterations.

2. Secondary users: Emergency Responders - 20%

    (a) Emergency responders represent an important but situational user group. While their use of SubterraView is less frequent than that of construction workers and utility technicians, the consequences of system failure during an emergency response are significant. Their requirements are given strong consideration but take lower priority than those of key users when conflicts arise.

3. Unimportant users: System Administrator- 10%

    (a) System administrators interact with SubterraView primarily through backend data management tools rather than the core application interface. Their requirements are given the lowest design priority, though their role in maintaining data accuracy is still essential to overall system reliability.

6 Mandated Constraints

6a Solution Constraints

1. 811 Services:

    (a) Description: 811 is a federally mandated call-before-you-dig service that is contacted before digging takes place. SubterraView will abide by this mandate and incorporate it into the application. SubterraView shall incorporate reminders and guidance for users to contact 811 services before any excavation begins, in accordance with federal mandate.

    (b) Rationale: SubterraView is designed to supplement the 811 processes, not replace it. All excavation activity must comply with federal call before you dig requirements regardless of what the app displays.

    (c) Fit Criterion: SubterraView shall display a prompt directing Construction Workers to contact 811 before excavation begins on any new project site.

2. Data Accuracy and Validation:

   (a) Description: SubterraView shall use an AI engine to cross reference data from multiple sources including public utility records, city infrastructure databases, and crowdsourced reports. The system shall assign a confidence rating to all infrastructure data and shall not present assumed or unverified information as confirmed.

   (b) Rationale: The reliability of the AR infrastructure overlay depends entirely on the accuracy of the underlying data. Displaying inaccurate infrastructure locations could result in utility strikes or worker injuries.

   (c) Fit Criterion: The AI engine shall flag any data that falls below the minimum confidence threshold and shall clearly indicate to the user when data in a given area is unverified or potentially outdated.

3. Platform and Hardware Compatibility:

   (a) Description: SubterraView shall be compatible with both iOS and Android smartphones. External hazard detection sensors shall connect to the app via Bluetooth and shall meet minimum weight and size requirements so as not to interfere with standard field equipment.

   (b) Rationale: Construction Workers and Utility Technicians operate in field environments where a lightweight, smartphone-based solution is the most practical and accessible option. Modern smartphones do not have built in gas or hazard detection capabilities, making minimal external hardware necessary.

   (c) Fit Criterion: SubterraView shall be approved for deployment on both iOS and Android platforms. External sensors shall weigh no more than 500 grams and shall pair with the app via Bluetooth without requiring additional software installation.

6b Implementation Environment of the Current System

SubterraView will operate primarily in outdoor field environments where Construction Workers, Utility Technicians, and Emergency Responders access the application through iOS and Android smartphones. The app requires an active internet connection to retrieve infrastructure data from the City Infrastructure Database and Utility Database in real time. In areas with limited connectivity, the system shall cache recently accessed infrastructure data locally on the device to maintain basic functionality.

External Bluetooth sensors will be deployed at job sites to detect hazardous underground conditions such as gas leaks and structural instabilities. These sensors will communicate with the SubterraView application wirelessly and trigger hazard alerts through the Hazard Monitoring System. The app will rely on the device camera for AR visualization and GPS for precise location mapping. All data processed by the system will pass through the AI validation engine before being displayed to the user.

6c Partner or Collaborative Applications

1. 811 Service Applications

    (a) SubterraView must interface with the 811-service notification system to ensure users are directed to complete the required call before you dig process before excavation begins. This integration supports compliance with federal excavation safety requirements and ensures SubterraView operates as a supplement to the existing 811 process rather than a replacement.

2. Hazard Sensor Network

    (a) SubterraView must communicate with external hazard detection sensors via Bluetooth to receive real time environmental readings. The app must be capable of processing incoming sensor data and triggering hazard alerts through the Hazard Monitoring System within a timeframe that meets OSHA worker protection requirements.

3. GIS Platform

    (a) SubterraView must integrate with an established geospatial platform to support the mapping and AR visualization functions of the application. This integration allows SubterraView to accurately position underground infrastructure data relative to real world coordinates and ensures that the infrastructure map overlay aligns correctly with the physical environment as viewed through the smartphone camera.

6d Off-the-Shelf Software

1. Cloud Services (Google Cloud Platform or Amazon Web Services) - SubterraView will use an established cloud platform to host the AI validation engine, store infrastructure data, and manage real time data transfers between the app and external databases. These platforms provide the scalability and reliability needed to support multiple simultaneous users across different job sites.

2. AI/ML frameworks (TensorFlow or PyTorch) - SubterraView will use an established machine learning framework to build and deploy the AI engine responsible for cross referencing infrastructure data and assigning confidence ratings. These frameworks provide prebuilt tools for model training and data validation that reduce development time.

3. GIS/GPS Software (Esri ArcGIS or Google Maps Platform) - SubterraView will integrate with an established geospatial platform to support real time location mapping and AR overlay functionality. These platforms provide the coordinate systems and mapping APIs necessary for accurately positioning underground infrastructure data relative to the physical environment.

4. Databases (PostgreSQL with PostGIS extension) - SubterraView will use a relational database system with geospatial support to store and manage

infrastructure records, confidence ratings, and crowdsourced reports. PostGIS extends PostgreSQL with geographic data types, making it well suited for storing and querying location-based infrastructure data.

5. AR Development Framework (ARKit (iOS) and ARCore (Android)) - SubterraView will use Apple ARKit and Google ARCore to power the augmented reality visualization features of the application. These are the native AR development frameworks for their respective platforms and are required to ensure stable AR performance across supported devices.

6e Anticipated Workplace Environment

SubterraView will be used primarily in outdoor environments including active construction sites and utility work zones. These environments present several design challenges that the product must account for. Construction sites are typically loud, with heavy machinery operating nearby, so SubterraView will use visual alerts and phone vibration in addition to audio notifications to ensure hazard warnings reach the user regardless of ambient noise levels. Outdoor use also means the display must remain readable in direct sunlight, requiring high contrast interface design. External hardware sensors must be weather resistant to function reliably in rain or dusty field conditions. Since Construction Workers and Utility Technicians may be wearing gloves or operating with limited hand mobility, the interface must support simple touch interactions and minimize the need for precise input.

6f Schedule Constraints

SubterraView estimates a total development timeline of approximately eighteen months. The first twelve months will cover system design, AI engine development, database integration, and AR feature implementation. The following six months will be dedicated to structured testing, bug fixes, and initial feedback collection from a selected group of Construction Workers and Utility Technicians before the full release on iOS and Android platforms.

Meeting this timeline is critical because the underground utility mapping market is growing rapidly and delays in release could result in competing products capturing the target customer base first. If the 18-month deadline is not met, the development team will prioritize core features including AR visualization, AI data validation, and hazard alerting, and defer lower priority features to a post launch update cycle rather than delay the full release.

6g Budget Constraints

The budget for SubterraView must account for several major cost areas including cloud infrastructure, AI engine development and training, GIS and mapping platform licensing, external sensor hardware development, and mobile application development for both iOS and Android. Given that SubterraView is being developed as a student project, the initial budget is limited, and the development team will prioritize the use of open-source frameworks and free tier cloud services where possible to reduce costs during the development phase.

A realistic minimum budget for a production ready version of SubterraView would need to cover ongoing API licensing fees, cloud hosting costs, and hardware prototyping. If funding falls short of what is required to build and maintain all planned features, the team will focus resources on the core AR visualization and hazard alerting functions, as these represent the highest priority requirements for the primary user groups.

7 Naming Conventions and Definitions.

7a Definitions of Key Terms

Augmented Reality: A technology that overlays digital information and visual elements onto the real-world environment using devices such as smartphones. In SubterraView, augmented reality is used to display underground infrastructure directly over a live camera view.

Artificial Intelligence: A computational system that takes large datasets, detects patterns, and makes decisions or predictions based on the information given. In SubterraView, AI is used to cross-reference underground infrastructure data from multiple sources, assign reliability scores, and update the infrastructure model.

Subterranean Infrastructure: Underground systems that support municipal and utility operations, including water pipelines, sewage systems, gas lines, electrical conduits, and telecommunications cables.

AI Validation Engine: An AI component that verifies infrastructure data by comparing records from utility databases, city infrastructure databases, and crowdsourced reports. It assigns confidence ratings and resolves conflicting data before updating the infrastructure map.

Confidence Rating: A numerical or categorical score generated by the AI system that represents the reliability and verification level of mapped infrastructure data.

Crowdsourcing: The process of collecting information from field workers or system users who submit reports about infrastructure discrepancies, new utilities, or hazard observations through the SubterraView application.

Sensor Units: Hardware devices that detect underground hazards such as gas leaks, structural instability, or abnormal environmental conditions and transmit alerts to the SubterraView system.

Hazard Monitoring System: A system module that receives input from sensor units and analyzes environmental risks to provide real-time safety alerts to workers and emergency responders.

7b UML and Other Notation Used in This Document

This document generally follows the UML 2.0 standard for all diagrams and notations. Any exceptions or custom symbols are noted below.

1. Interaction Diagram (Figure 1)
   - (a) Symbol: Large white circle in the center
     - i. Meaning: Core system functions
   - (b) Symbol: Black circles at the edges
     - i. Meaning: External users, interacting with the system
   - (c) Symbol: Solid black arrows
     - i. Meaning: What the users give to the system and vice versa
2. Use Case Diagram (Figure 2)
   - (a) Symbol: Circle on top of an open semicircle
     - i. Meaning: External user that interacts with the system
   - (b) Symbol: Database cylinder
     - i. Meaning: External data that interacts with the system
   - (c) Symbol: Alarm
     - i. Meaning: Hazard Detection System
   - (d) Symbol: Solid arrow between an external role and use case
     - i. Meaning: Interactions/Initiates use case

7c Data Dictionary for Any Included Models

Variable Name	**Data Type**	**Description**
InfrastructureID	String	Unique identifier assigned to each infrastructure element, 12 characters, alphanumeric

InfrastructureType	String	Type of utility infrastructure (water, gas, electric, telecommunications, sewage), only English words
GeographicLocation	Tuple(Longitude, Latitude)	Location of underground infrastructure, doubles
ConfidenceRating	Float	Reliability score assigned to infrastructure data, 0.0-1.0
HazardLevel	Integer	Risk level detected by sensor units, 0-10
ReportID	String	Unique identifier for crowdsourced infrastructure reports, 12 characters, alphanumeric
SensorID	String	Unique identifier assigned to each hazard detection sensor, 12 characters, alphanumeric
Timestamp	DateTime	Date and time when data was recorded or updated
UserID	String	Identifier for the worker or responder interacting with the system, 12 characters, alphanumeric
MapOverlayData	AR Object Data	Visual infrastructure model displayed through augmented reality

Infrastructure image data is stored in JPEG or PNG format, telemetry data is stored in CSV format, and AR overlay data is stored in GLTF format.

8 Relevant Facts and Assumptions

8a Facts

Underground infrastructure management currently relies heavily on the 811 call-before-you-dig notification system.

The 811 process takes multiple days and relies on field markings.

Nearly 197,000 underground utility damage incidents were reported in 2024 according to the Common Ground Alliance DIRT Report.

Underground utility damage incidents cost an estimated $30 billion annually in societal and economic losses.

SubterraView depends on data provided by utility companies and municipal agencies.

Hazard monitoring depends on sensor units detecting gas leaks, structural risks, or environmental abnormalities.

The system supports construction planning, excavation safety, and emergency response coordination.

SubterraView does not perform physical excavation, infrastructure maintenance, or regulatory enforcement.

8b Assumptions

Utility companies and municipal agencies will allow access to infrastructure data.

Users will receive training to operate the SubterraView application.

Reliable internet connectivity will usually be available during system operation.

AI validation will maintain approximately 90% infrastructure data accuracy.

Infrastructure repair and regulatory compliance responsibilities remain with external organizations.

II Requirements

SV: Sections 9 and 10 deal with functional requirements. Sections 11 to 20 are a very thorough list of possible non-functional requirements, not all of which apply to every project. You should think carefully about each of these, form requirements if applicable, or write "Not Applicable" otherwise. See section 10 for the format of individual requirements. Section 21 documents the acceptance tests planned to verify the requirements – See that section for further details, and be aware that every requirement needs at least one verifying acceptance test (though some tests may verify more than one requirement.)

9 Product Use Cases

SV: Product Use Cases are very similar to Product Scenarios, but in more formal detail. They serve as a first step towards developing functional requirements, and can aid in organizing requirements according to the use case(s) from which they were developed. See the CS 440 web site for a sample use-case form, with instructions.

9a Use Case Diagrams

SV: Use case diagrams list the use cases developed for a system, mark the boundary of what is internal or external to the system to be developed, and indicate which external entities (actors) are associated with each use case.

Examples

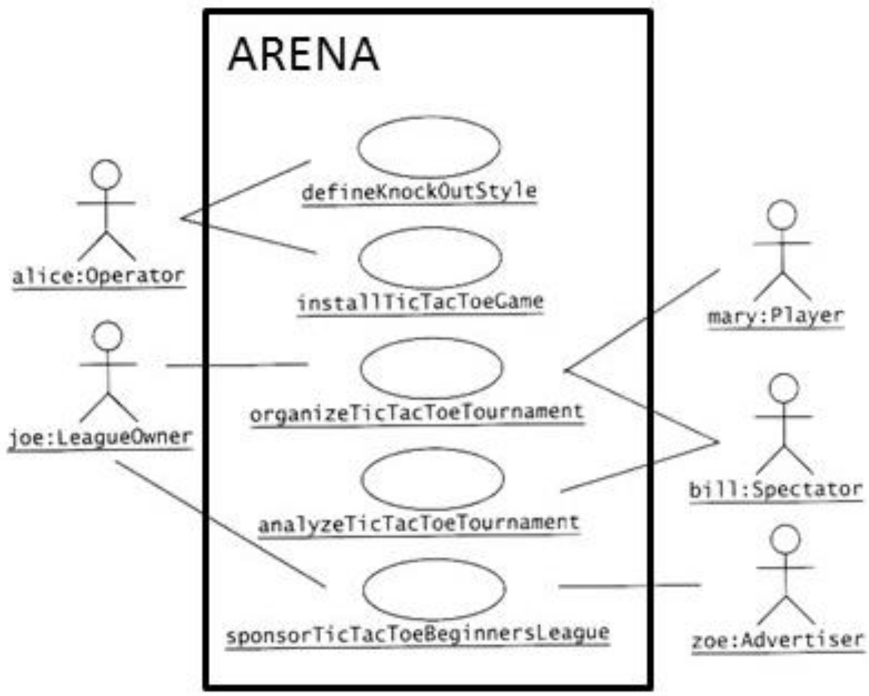


Figure 3 – Sample Use Case Diagram from Bruegge & DuToit (modified)

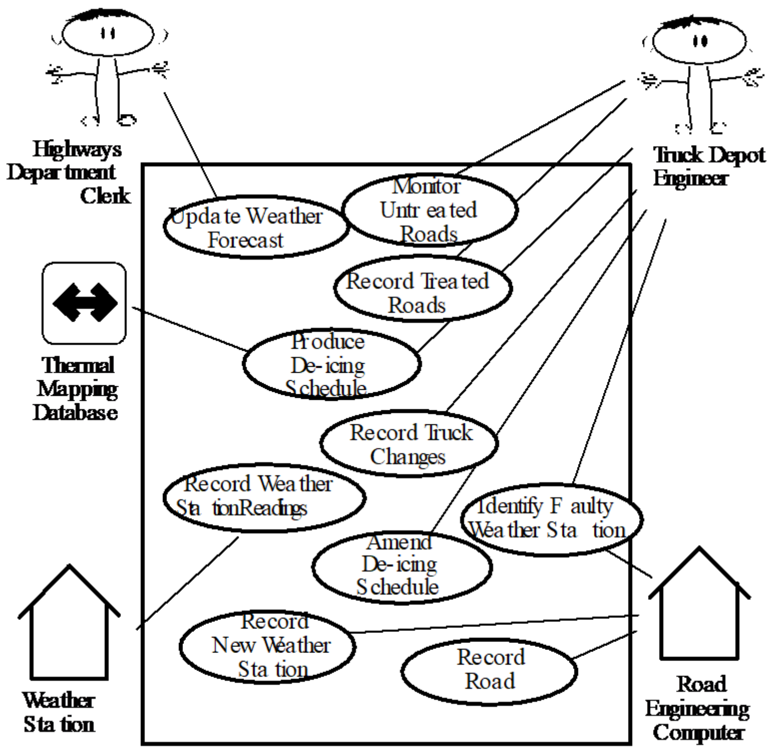


Figure 4 - Sample Use Case Diagram from Robertson and Robertson

9b Product Use Case List

SV: A list (table) of use cases is an alternative to the use case diagram, particularly when there are many use cases. There may be additional information in the table not found in the diagram, such as cross referencing to other sections or materials.

9c Individual Product Use Cases

SV: The following example was copied from "useCaseFormWithInstructions.docx", available on the CS 440 web site. (There is also a blank version available.)

Use case ID: Name: pre-conditions: post-conditions: Initiated by: Triggering Event: Additional Actors:

Sequence of Events:
 	
1. Initiating event or action should be step 1, taken by initiating actor.
    2. System response follows, indented right.
3. All external action steps are aligned with step 1. ("stimulus" style)
    4. All system responses are indented right, aligned with step 2. ("response" style)
5. All steps should be expressed in the active voice, clearly indicating **<u>who</u>** performs each action
    6. The sequence of events should show a back-and-forth stimulus-response relationship.

Alternatives: These would be normal and expected variations from the base case. Exceptions: These would be unusual variations from the base case, often caused by problems.

- *For all of the above, list as NA if not applicable.*
- *The following may be added if relevant, or omitted otherwise:*
  - *related use cases or scenarios*
  - *associated tests, systems, classes, etc.*
  - *revision history*
  - *references to other documents*
  - *author(s) / originator(s)*

    - *notes*
- *Alternatives and Exceptions may be listed either as separate use cases or as notes to a base case, depending on their significance and similarity.*
- *For regularly occurring periodic events, "time" can be listed as the initiating actor.*

10 Functional Requirements

SV: Each requirement listed needs to have a unique identifier, a short name, a one- or two-sentence description, a rationale, a fit criteria, and reference to one or more acceptance tests to be used to confirm the completion of this particular requirement. The acceptance tests themselves are documented in section 0- See that section for further details. It is recommended to number the requirements according to their type, such as F-4 for the fourth functional requirement or U-2 for the second usability requirement. Functional requirements specifically deal with the functionality the system must have, and are generally derived directly from the steps the system takes during use cases.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

11 Data Requirements

SV: Data requirements deal with requirements that are somehow related to data, such as the definition of what is included in a "student record" or the acceptable form of an e-mail address or allowable range of certain data items.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

12 Performance Requirements

12a Speed and Latency Requirements

SV: Requirements specifying how fast (or slow) the product must operate or how much lag is allowable between stimulus and either initial response or task completion. Other timing-related requirements could go in this section.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

12b Precision or Accuracy Requirements

SV: Self-explanatory. How accurate or precise must the system be.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

12c Capacity Requirements

SV: Requirements regarding the largest "thing" the system must be able to handle, or perhaps how many things it can handle (at once.) Note: Requirements regarding how many things it can handle in a given time period would be a speed requirement, covered in section 12a above.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

13 Dependability Requirements

13a Reliability Requirements

SV: Reliability relates to how frequently the system fails, (either by shutting down or by delivering erroneous results), and the consequences of those failures. These requirements may also address the conditions under which it is allowed to fail (or not.), See also availability and robustness in the following sections.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

13b Availability Requirements

SV: Availability addresses the amount of time the system is running and available for use. It is affected by how often the system goes down (reliability), but also by the time required to bring the system back up again, the availability lost due to regularly scheduled maintenance down times, and the ability of the system to offer at least partial functionality in the face of failures or resource shortages. See also reliability and robustness.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

13c Robustness or Fault-Tolerance Requirements

SV: This section deals with the system's ability to provide at least partial functionality in the face of failures or resource shortages, such as operating in offline mode when network connectivity is unavailable. See also reliability and availability.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

13d Safety-Critical Requirements

SV: These requirements address potential harm to health, safety, or property, and may refer to relevant standards such as OSHA compliance.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

14 Maintainability and Supportability Requirements

14a Maintenance Requirements

SV: This section deals with the ease with which the system can be maintained, and possibly who will perform system maintenance and under what conditions. The ease of evolving the system into future versions may also be addressed here, or in a separate section (not included in this template) if that is a major concern.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

14b Supportability Requirements

SV: What ongoing support is to be provided, e.g. through a help desk. See also training requirements in section 16g below.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

14c Adaptability Requirements

SV: Description of other platforms or environments to which the product must be ported.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

14d Scalability or Extensibility Requirements

SV: The ease of expanding the system to a larger capacity as the business grows.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

14e Longevity Requirements

SV: This specifies the expected lifetime of the product.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

15 Security Requirements

SV: Security requirements address who is allowed what type of access to the system, and what areas require special protection or diligence. In practice security requirements must often be written by security experts, and may refer to standards.

15a Access Requirements

SV: These requirements address who has access to what (data or functionality) and under what conditions or restrictions.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

15b Integrity Requirements

SV: These requirements address the protection of data(bases) from intentional or accidental corruption, loss, or theft.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

15c Privacy Requirements

SV: These requirements address data that must remain confidential, such as medical records or other personally identifiable data. Laws often apply. (See also section 20.)

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

15d Audit Requirements

SV: This section applies when a system must provide support for transaction auditing, such as some financial or medical systems.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

15e Immunity Requirements

SV: This section addresses the system's ability to resist viruses, worms, Trojan Horses, etc.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

16 Usability and Humanity Requirements

SV: This section is concerned with requirements that make the product usable and ergonomically acceptable to its hands-on users.

16a Ease of Use Requirements

SV: This section addresses the ease with which the intended audience can use the system properly, and conversely the difficulty with which they can use it improperly.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

16b Personalization and Internationalization Requirements

SV: This section addresses the ease with which the system can be configured for personal preferences, and for things such as language, currency, units, symbols, etc.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

16c Learning Requirements

SV: Requirements related to how easy it is for the intended audience to learn to use the product.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

16d Understandability and Politeness Requirements

SV: These requirements relate to how intuitively the intended audience understands what the program does, what its messages mean, and how to use it. Definitely related to ease of use, (section 16a), but more specifically addressing comprehension of the program output, instructions, and other messages.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

16e Accessibility Requirements

SV: Requirements related to use of the product by individuals with disabilities.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

16f User Documentation Requirements

SV: List of the user documentation to be supplied as part of the product.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

16g Training Requirements

SV: A description of the training needed by users of the product.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

17 Look and Feel Requirements

17a Appearance Requirements

SV: These requirements address things such as the colors, fonts, and logos used, often to reflect corporate branding or similarity to related products. See also style in the next section.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

17b Style Requirements

SV: Style requirements address the impression the product makes upon users, such as professionalism for a tax accounting package, friendliness for a children's game, or how "cool" it is for a teenage audience. Product packaging may also be addressed here, and/or appearance in the previous section.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

18 Operational and Environmental Requirements

18a Expected Physical Environment

SV: These requirements relate to the physical environment in which the product will operate.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

18b Requirements for Interfacing with Adjacent Systems

SV: This section describes the requirements to interface with partner applications and/or devices that the product needs to successfully operate.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

18c Productization Requirements

SV: Requirements related to the distribution and/or installation of the product.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

18d Release Requirements

SV: Specification of the intended release cycle for the product and the form that the release shall take.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

19 Cultural and Political Requirements

19a Cultural Requirements

SV: This section contains requirements that are specific to the sociological factors that affect the acceptability of the product. If you are developing a product for foreign markets, then these requirements are particularly relevant. Bear in mind that "cultural groups" may also apply to population subgroups such as teenagers, the elderly, or ironworkers.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

19b Political Requirements

SV: Requirements included strictly to make "the boss" happy, either internally to the development company, or internally to the client company, or possibly an external third party.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

20 Legal Requirements

20a Compliance Requirements

SV: A statement specifying the legal requirements for this system, often referring to relevant laws and/or requiring approval by the legal department.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

20b Standards Requirements

SV: These requirements specify documented standards to which the product must conform, as opposed to legal regulations.

ID# - Name

Description: Your description here . . .

Rationale: Your rationale here . . .

Fit Criterion: Your fit criteria here . . .

Acceptance Tests: List ID# and/or names here . . .

21 Requirements Acceptance Tests

SV: Every requirement must have one or more acceptance tests associated with it, to confirm that the requirement has been met. At this point these tests are not yet completely specified – A one- or two-sentence description of each test will suffice. Note that some tests may verify more than one requirement, and that some requirements may require multiple tests for their confirmation.

21a Requirements – Test Correspondence Summary

SV: The following sample table is available from the CS 440 web site as "Sample Requirement Test Correspondence Table.xlsx" It is recommended that you work with the table in Excel, and then drag it into the document when it is completed. Depending on the number of requirements and/or tests included, it may be necessary to use multiple tables, and/or use landscape mode. Every row and every column of the table should include at least one X. Below the table list the ID #, name, and short description of each individual acceptance test.

	Requirements																			
Test	Req 1	Req 2	Req 3	Req 4	Req 5	Req 6	Req 7	Req 8	Req 9	Req 10	Req 11	Req 12	Req 13	Req 14	Req 15	Req 16	Req 17	Req 18	Req 19	Req 20
Test 1	X																			
Test 2		X				X														
Test 3			X	X																
Test 4					X	X														
Test 5																				
Test 6																				
Test 7																				
Test 8																				
Test 9																				
Test 10																				
Test 11																				
Test 12																				
Test 13																				
Test 14																				
Test 15																				

Table 1 - Requirements - Acceptance Tests Correspondence

21b Acceptance Test Descriptions

SV: Provide a brief description of each acceptance test. Detailed test specifications will appear in a separate document, which may be referenced here when available.

ID # - Name

Description: Your description here . . .

III Design

22 Design Goals

SV: Identify the important design goals that are to be optimized in the proposed design.

Your text goes here . . .

23 Current System Design

*SV: __**IF**__ the proposed new system is to replace an existing system, then the current system should be described here. Otherwise insert a brief statement that there is no pre-existing system.*

Your text goes here . . .

24 Proposed System Design

This section will make heavy use of class diagrams, and also sequence and deployment diagrams where noted. However don't overlook finite state, activity, communication, or other diagram types as needed for effective communication.

24a Initial System Analysis and Class Identification

SV: Perform grammatical and similar analyses to identify the most import and obviously needed classes, and to organize them into an initial class structure. An initial class diagram is appropriate, containing few if any internal details.

Your text goes here . . .

24b Dynamic Modelling of Use-Cases

SV: Insert sequence diagrams of (at least the most important) use-cases, as a means of identifying other needed classes.

Your text goes here . . .

24c Proposed System Architecture

SV: Identify the Software Architecture to be applied to this project, such as Client-Server, Repository, MVC, etc., along with justification for the choice.

Your text goes here . . .

24d Initial Subsystem Decomposition

SV: A slightly more detailed class diagram, showing the classes identified in sections 24a, 24b, and 0 above, partitioned into subsystems. For each subsystem provide a brief description of the subsystem, including its key responsibilities. There should still be few if any internal details.

Your text goes here . . .

25 Additional Design Considerations

SV: The sections listed here do not need to be presented in the order given, and may not all be relevant for any particular project. Those that are relevant can help identify additional classes that are needed as a result.

25a Hardware / Software Mapping

SV: This is particularly important for distributed systems, such as those employing a client-server architecture. Use a deployment diagram to indicate which subsystems are mapped onto which piece(s) of hardware, and what communication subsystems need to be added to the system as a result.

Your text goes here . . .

25b Persistent Data Management

SV: Document the classes and perhaps subsystems necessary to store persistent data when the system shuts down, and to restore that data when the system starts back up again.

Reiterate key data structures and information as necessary for the understanding of this design phase. Refer the reader back to the data dictionary in section ***Error! Reference source not found.*** *to avoid undue repetition, while reviewing only the most relevant items here.*

Your text goes here . . .

25c Access Control and Security

SV: Identify the access control and security concerns for this system, and the new classes and/or subsystems that must be added to handle those concerns.

Your text goes here . . .

25d Global Software Control

SV: Identify the global software control concerns for this system, and the new classes and/or subsystems that must be added to handle those concerns.

Your text goes here . . .

25e Boundary Conditions

SV: Identify the boundary condition concerns for this system, and the new classes and/or subsystems that must be added to handle those concerns. In particular consider startup, shutdown (normal or abnormal), and the creation and/or maintenance of any configuration files, databases, or similar supporting data files.

Your text goes here . . .

25f User Interface

SV: Include a preliminary user interface design here, possibly as a rough sketch or other mockup, in order to identify additional classes needed to implement the interface.

Your text goes here . . .

25g Application of Design Patterns

SV: Any design patterns applied as a result of previous sections should have been addressed there, and identified as such at the time. Use this section to document only the additional design patterns that were not previously covered elsewhere. (If any.)

Your text goes here . . .

26 Final System Design

SV: Include here the final version of the overall system design, incorporating all the subsystems and classes added as a result of additional design considerations. Multiple diagrams may be needed, possibly starting with an overall package diagram showing all the different subsystems and the (important) classes contained within each one. Still not a lot of internal details.

Your text goes here . . .

27 Object Design

This section documents the internal details of each class, to the extent that they can be designed at this time. Included should be the class interfaces (public method signatures and responsibilities) and constraints. It is probably best to break this section up into subsections corresponding to subsystems as documented above, and/or by (Java) packages if those are designed. It may also be appropriate to address additional design pattern considerations here, but not to the point of being redundant of previous documentation.

Certain methods, such as simple getters, setters, and constructors are not always documented, unless there is something special about them such as in the Singleton or Factory Method design patterns.

27a Packages

SV: If the design involves assigning classes to packages (.e.g Java packages), then the packages to be created should be documented here.

Your text goes here . . .

27b Subsystem I

Your text goes here . . .

27c Subsystem II

Your text goes here . . .

27d etc.

Your text goes here . . .

IV Project Issues

28 Open Issues

SV: Issues that have been raised and do not yet have a conclusion.

Your text goes here . . .

29 Off-the-Shelf Solutions

SV: Discussion of products or components currently available that could either be incorporated into the new solution or simply used instead of developing (parts of) the new solution. The distinction between sections 35 a, b, and c is subtle, and not very important.

Your text goes here . . .

29a Ready-Made Products

SV: Products available for purchase that could be used either as part of a solution or instead of (a part of) a solution.

Your text goes here . . .

29b Reusable Components

SV: Similar to 35a, but for components such as libraries or toolkits instead of fully blown products.

Your text goes here . . .

29c Products That Can Be Copied

SV: Products that could legally be copied would typically be past projects developed by the same development group, provided there were no restrictions that would prevent their reuse.

Your text goes here . . .

30 New Problems

SV: The proposed new system certainly has its benefits, but it could also raise new problems. It is a good idea to identify any such potential problems early on, rather than being surprised by them later.

30a Effects on the Current Environment

SV: Could the new system have any adverse effects on the working environment, e.g. the way people do their jobs?

Your text goes here . . .

30b Effects on the Installed Systems

SV: Could the new system have any adverse effects on other hardware or software systems?

Your text goes here . . .

30c Potential User Problems

SV: Could the new system have any adverse effects on the users of the software? Could users possibly have a negative response to the new system?

Your text goes here . . .

30d Limitations in the Anticipated Implementation Environment That May Inhibit the New Product

SV: Are there any (physical) limitations in the expected environment that could inhibit the proposed product? (e.g. weather, electrical interference, radiation, lack of reliable power, etc.)

Your text goes here . . .

30e Follow-Up Problems

SV: Basically any other possible problems that could occur.

Your text goes here . . .

31 Migration to the New Product

SV: This section only applies when there is an existing system that is being replaced by a new system, particularly when data must be preserved and possibly translated / reformatted. Otherwise just write "Not Applicable" under section 38 and remove sections 38a and 38b.

31a Requirements for Migration to the New Product

SV: These are a list of requirements relevant to the migration procedures. For example a requirement that the two systems be run in parallel for a time until the client is satisfied with the new system and the users know how to use it.

Your text goes here . . .

31b Data That Has to Be Modified or Translated for the New System

SV: This section specifically addresses ***data*** *that must be preserved and/or translated / reformatted during the migration process.*

Your text goes here . . .

32 Risks

SV: Consideration of the potential risks that could cause the project to fail / underperform.

Your text goes here . . .

33 Costs

SV: An estimate of what it will cost to complete this project. Think not only in terms of dollars, but also time, resources, lost opportunities, etc.

Your text goes here . . .

34 Waiting Room

SV: This is a place to record ideas or wishes that will not be included in the current release of the product, but which might be worth reconsidering at a later date.

Your text goes here . . .

35 Ideas for Solutions

SV: When developing requirements only, it is not the role of the business analyst to dictate the implementation of the solution. However they can pass along any ideas they have here as suggestions to the developers. For CS 440 this report includes system and object design, so this section would make suggestions for implementation and *testing that would come after design, such as the use of a particular language, IDE, library, or other tools.*

Your text goes here . . .

36 Project Retrospective

SV: At the conclusion of the (CS 440) project, reflect back on what worked well and what didn't, and how the process could be improved in the future.

Your text goes here . . .

V Glossary

SV: The glossary is a more complete and inclusive dictionary of defined terms than that found in section I.7.a, the latter of which only covered the most important key terms needed to understand the report.

Your text goes here . . .

VI References / Bibliography

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VII Index

This section provides an index to the report. The sample below was generated using the "Mark Entry" and "Insert Index" items from the "Index" section on the "References" tab, and can be automatically updated by right clicking on the table below and selecting "Update Field". To remove marked entries from the document, toggle the display of hidden paragraph marks (the paragraph button on the "Home" tab), and remove the tags shown with XE in { curly braces. }